

ELE 475 / COS 475
Computer Architecture
Lecture 7: Superpipelining

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Agenda

- Motivation for Long Pipelines
- Case Study: Intel Pentium 4
- Challenges for Long Pipelines

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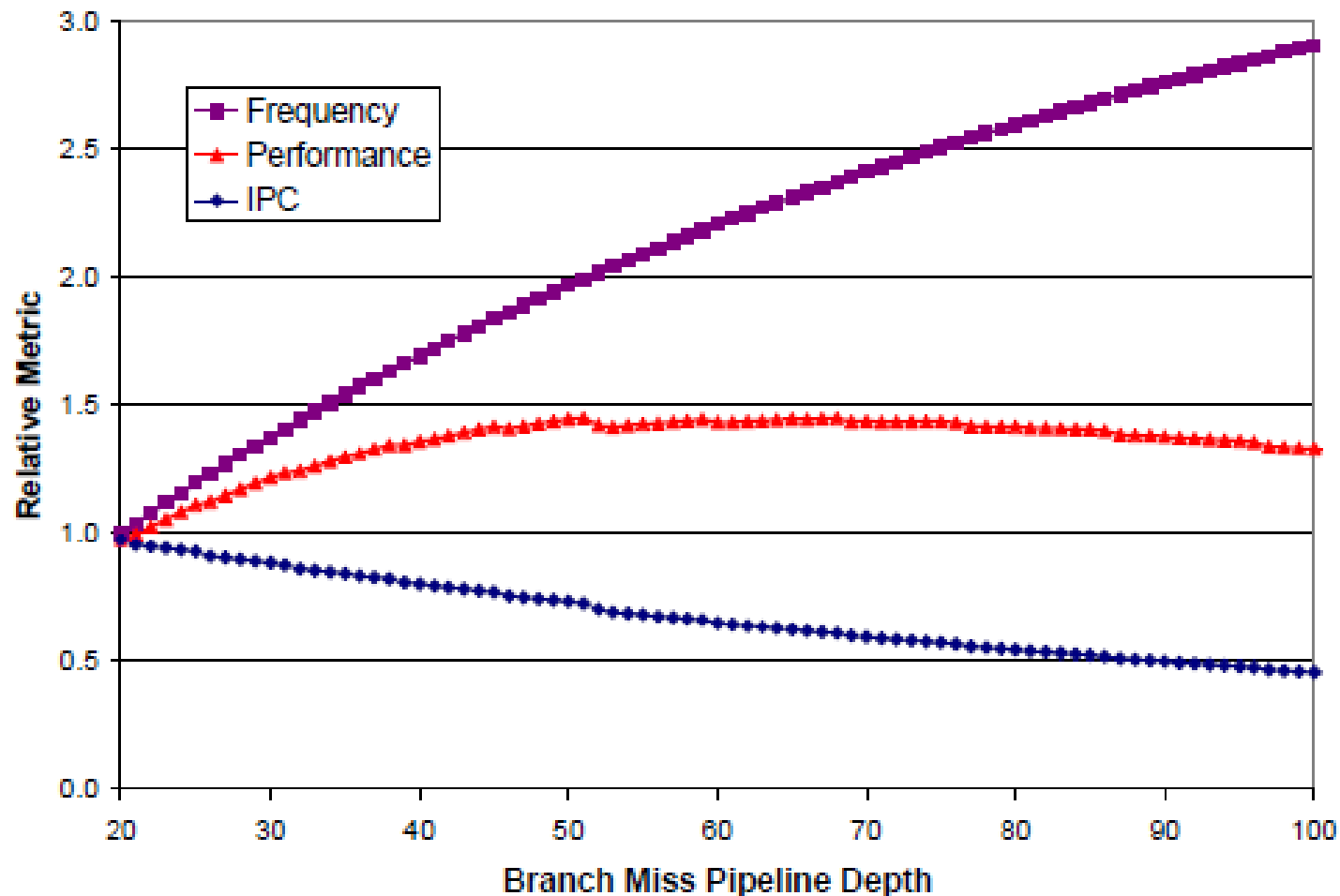
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Raising Clock Frequency

$$\frac{\text{Time}}{\text{Program}} = \frac{\text{Instructions}}{\text{Program}} * \frac{\text{Cycles}}{\text{Instruction}} * \frac{\text{Time}}{\text{Cycle}}$$

- Reducing cycle-time reduces program time
 - Increase pipeline depth
- Reducing CPI reduces program time
 - Wider issue/execute processors
- Reducing number of instructions reduces program time
 - Compiler optimization and complex instructions

Increasing Frequency Increases CPI



Workload Influences Optimal Pipeline Depth

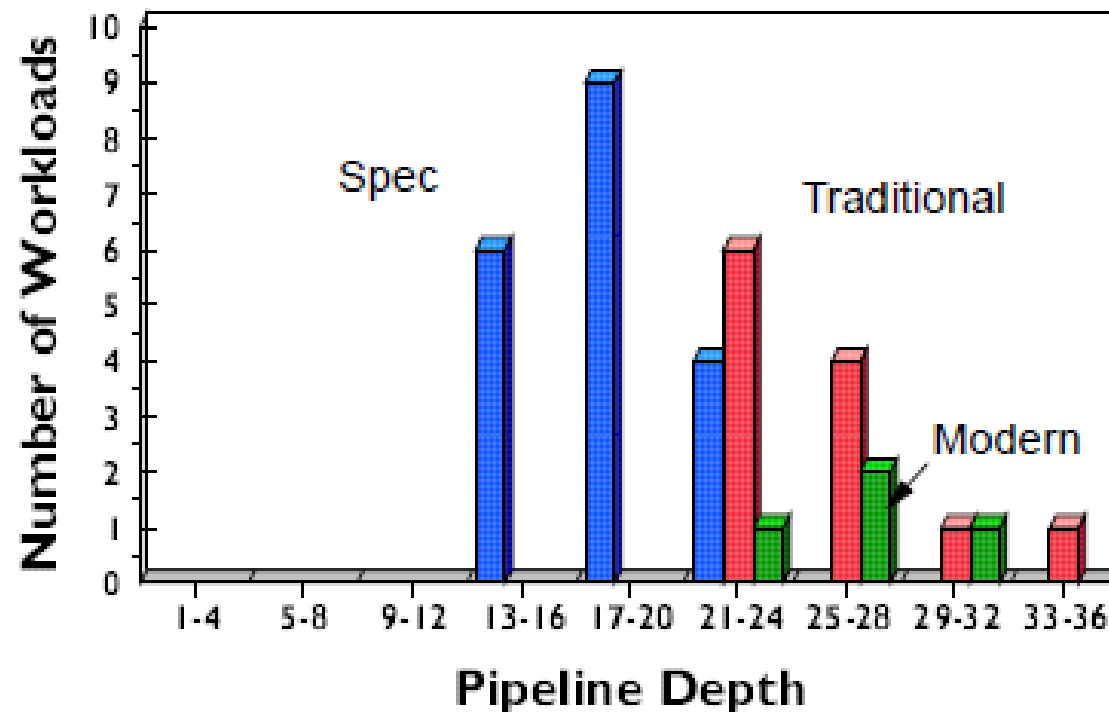


Fig. 9 shows the distribution of pipeline depth optima for different workload classes. Blue are the Spec (C) workloads, red are the traditional workloads, and green are the modern (C++ and Java) workloads.

Vector Processors and GPUs

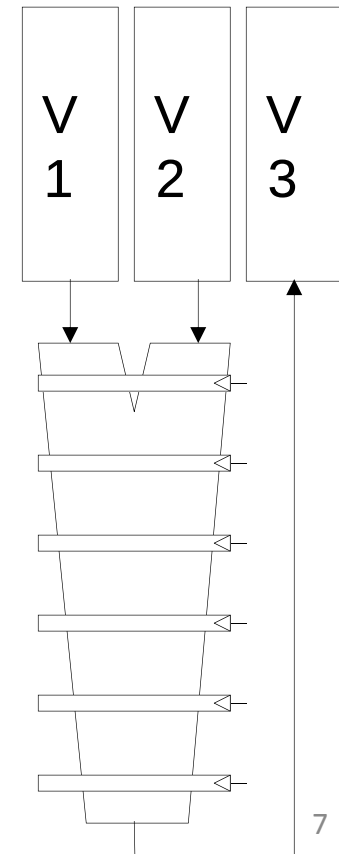
- Reduced interdependence of instructions
- Can create longer pipelines
- But tradeoff with parallel pipelines



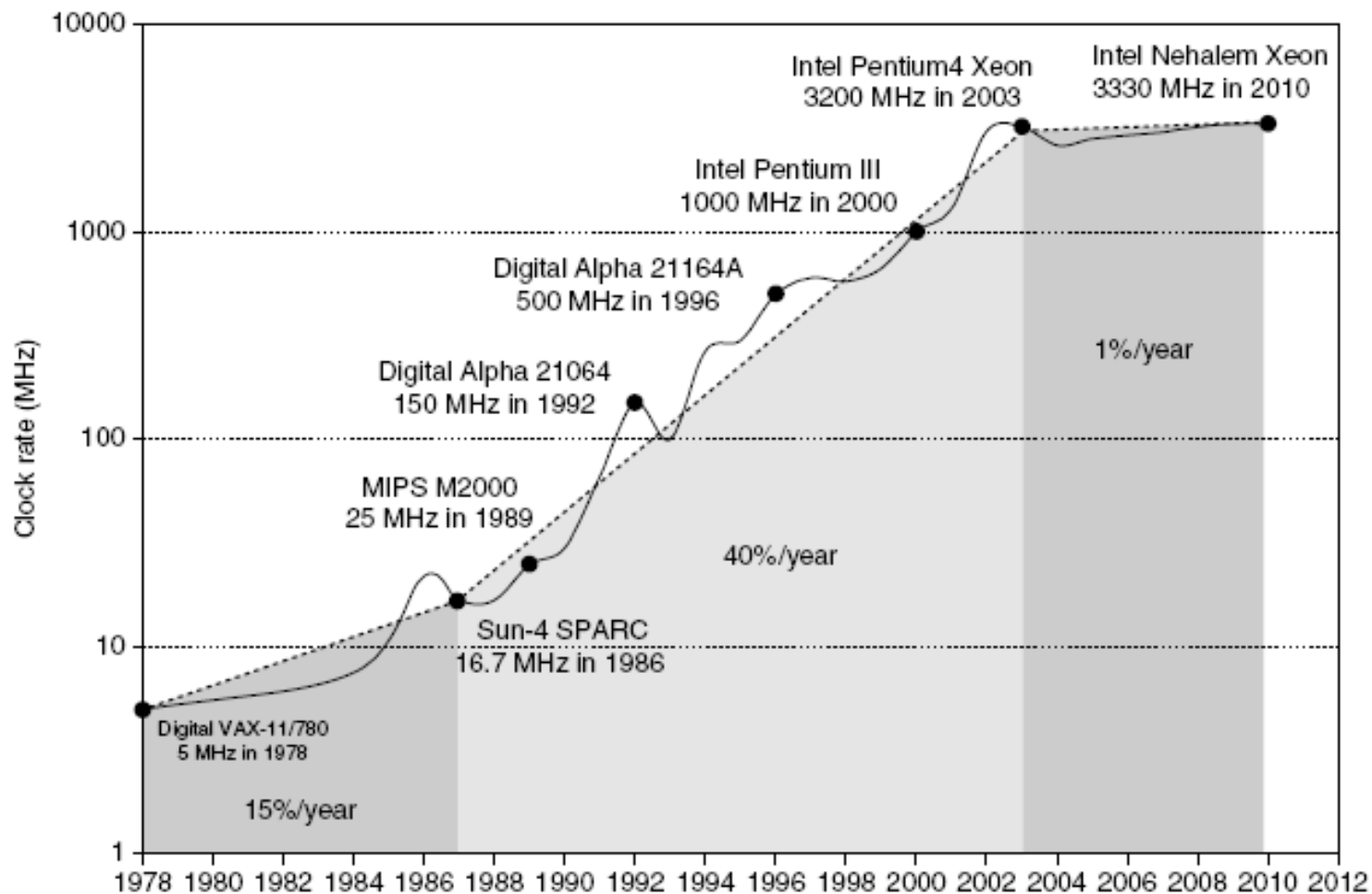
Cray 1

Cray 1 latencies:

Integer Add	3
Logical	2
Shift	4
FP Add	6
FP Mult	7
FP Recip (Div)	14



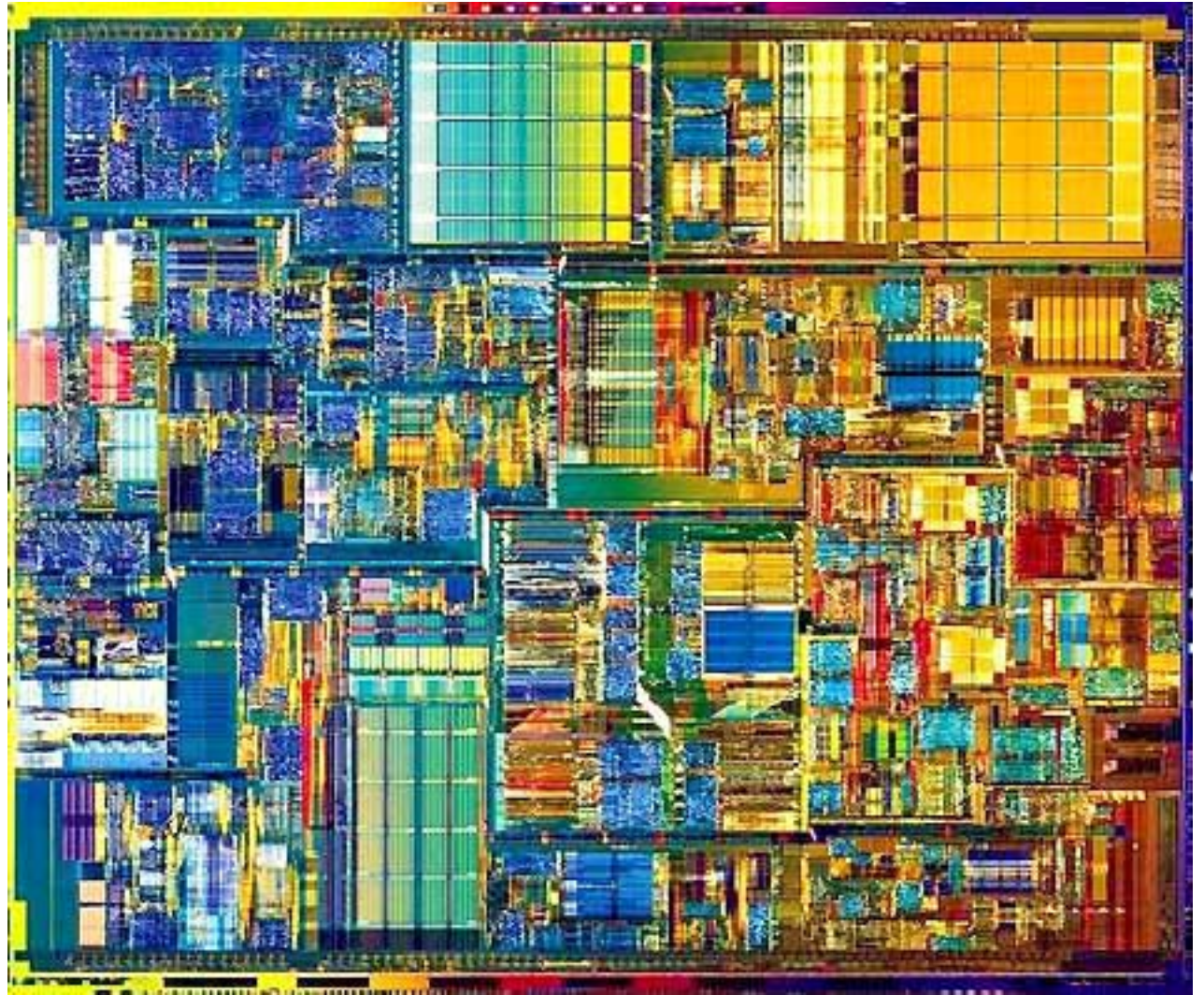
Clock Frequency Improvement



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Pentium 4



Pentium III vs. Pentium 4 Pipeline

Basic Pentium III Processor Misprediction Pipeline

1	2	3	4	5	6	7	8	9	10
Fetch	Fetch	Decode	Decode	Decode	Rename	ROB Rd	Rdy/Sch	Dispatch	Exec

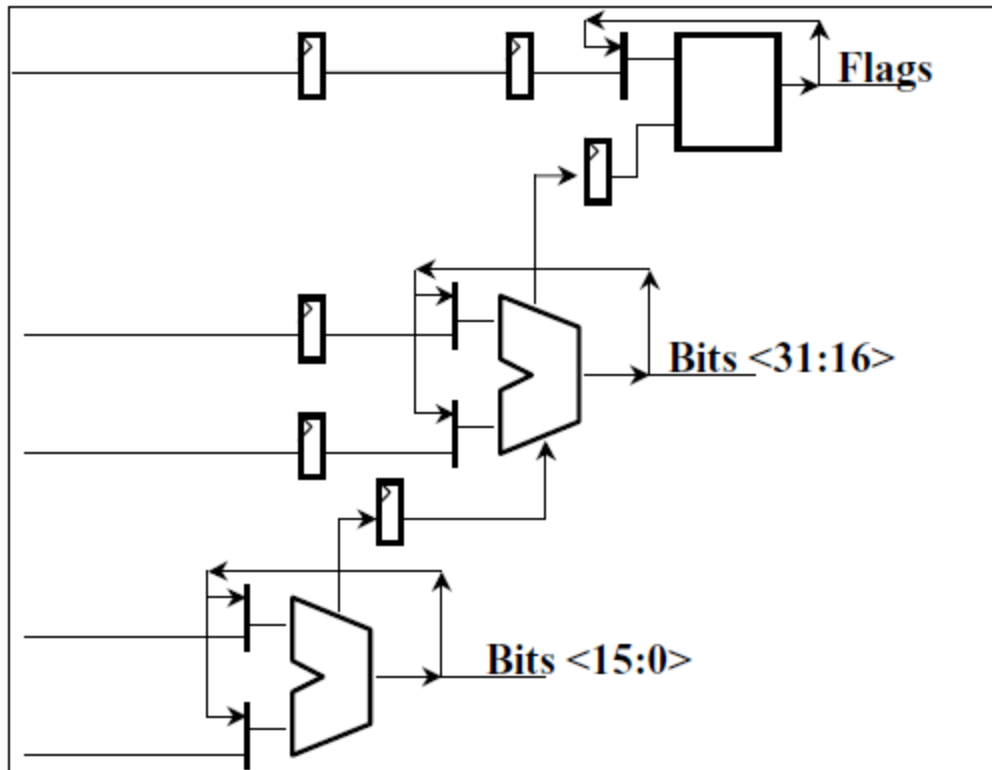
Basic Pentium 4 Processor Misprediction Pipeline

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
TC Nxt IP		TC Fetch		Drive	Alloc	Rename		Que	Sch	Sch	Sch	Disp	Disp	RF	RF	Ex	Flgs	Br Ck	Drive

Many more stages...

- Drive Stage needed just to drive signals between pipelines

Double Pumped ALU

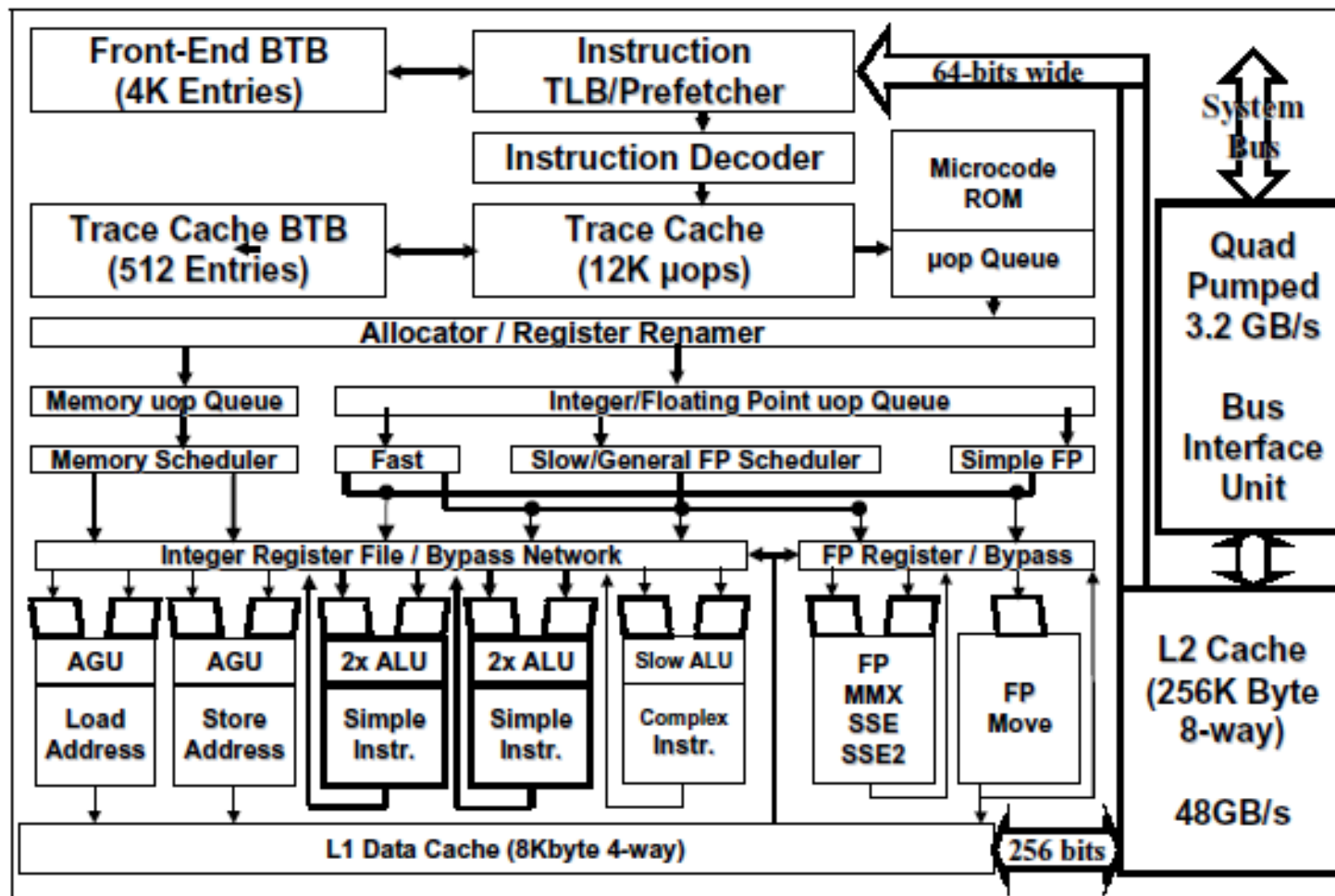


- Executes at 2x core frequency!
- Bypass low bits early within pipeline
- Cut carry of 32-bit add
- Calculate Flags in 3rd cycle

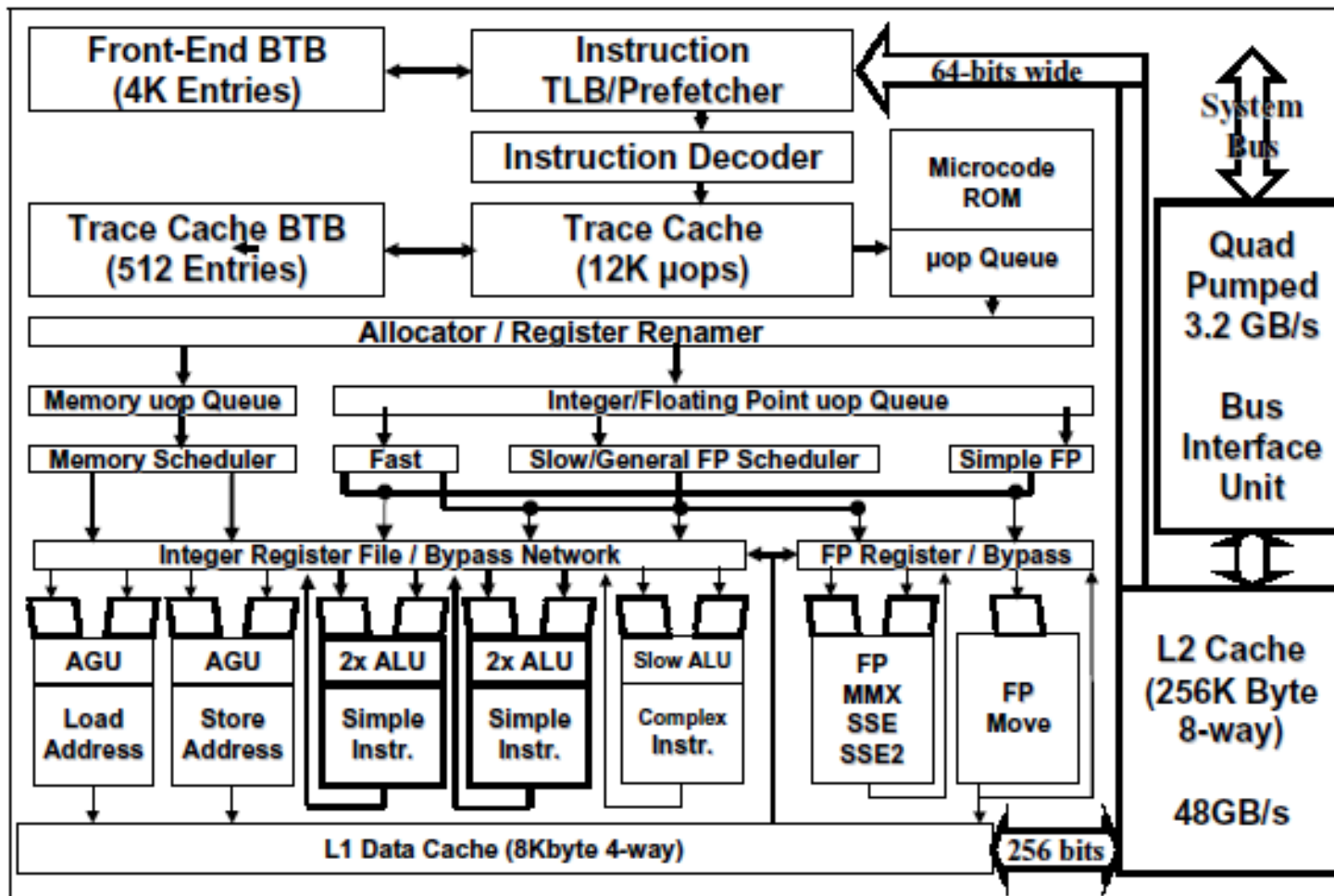
Small L1 Cache for Low Latency

- 8KB L1 Cache
- Much smaller than contemporary L1 caches
- Done to allow low latency of 2-cycles.
- Write-through to L2
- Low latency cache especially important to x86 ISA because so few registers requires going to stack often

Trace Cache



Pentium 4 Architecture



Other Deeply Pipelined Processor

Really Deep:

- Power 7

Deep:

- Intel Core
- Intel Core 2
- Intel Core i3, i5, i7 (Nehalem etc.)
- Intel Core i3, i5, i7 2nd Gen. (Sandy/Ivy Bridge)

Agenda

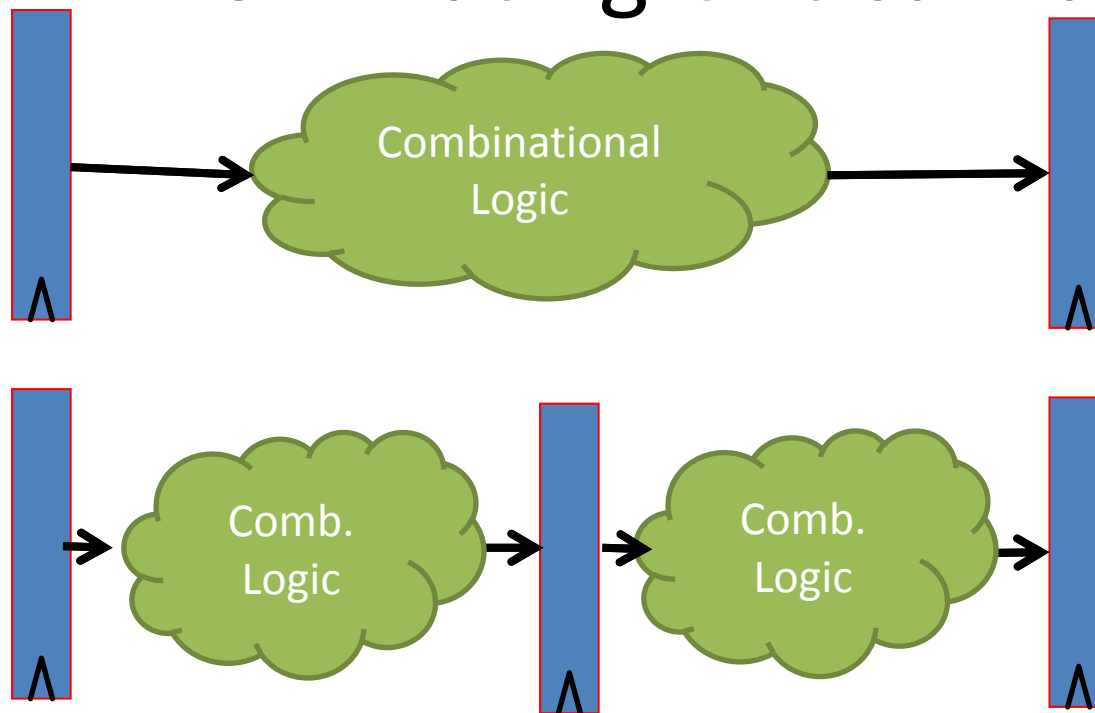
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Problems with Superpipelining

- Hard to build
- Hard to cut critical paths
- More bypassing needed
- More windows
- Flip-flop overheads become larger part of cycle
- Large pipelining requires more “pipeline” parallelism in program
 - Requires better branch prediction
- Power!!!

Flip Flops/Latches

Dominating Critical Path

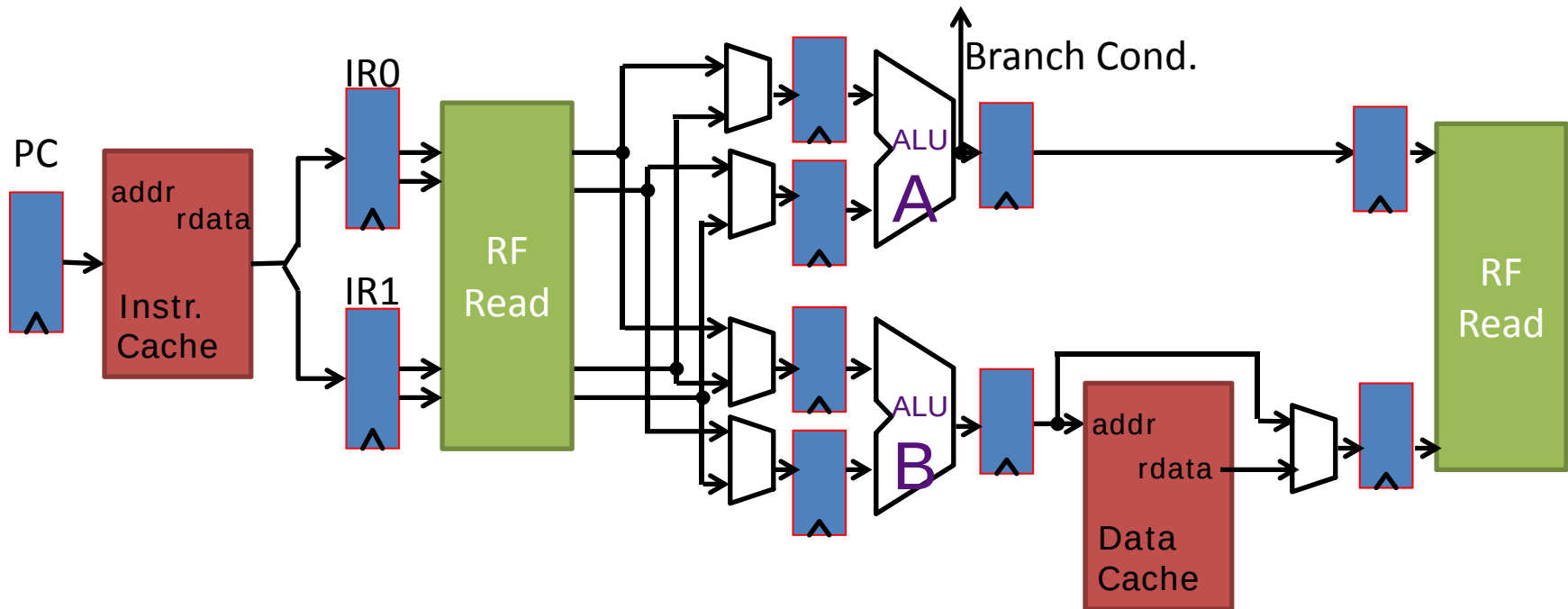


- Flip Flop/Latch delay becomes larger portion of cycle
- Setup Time Dominates with short cycles
- Can also have Hold Time problems

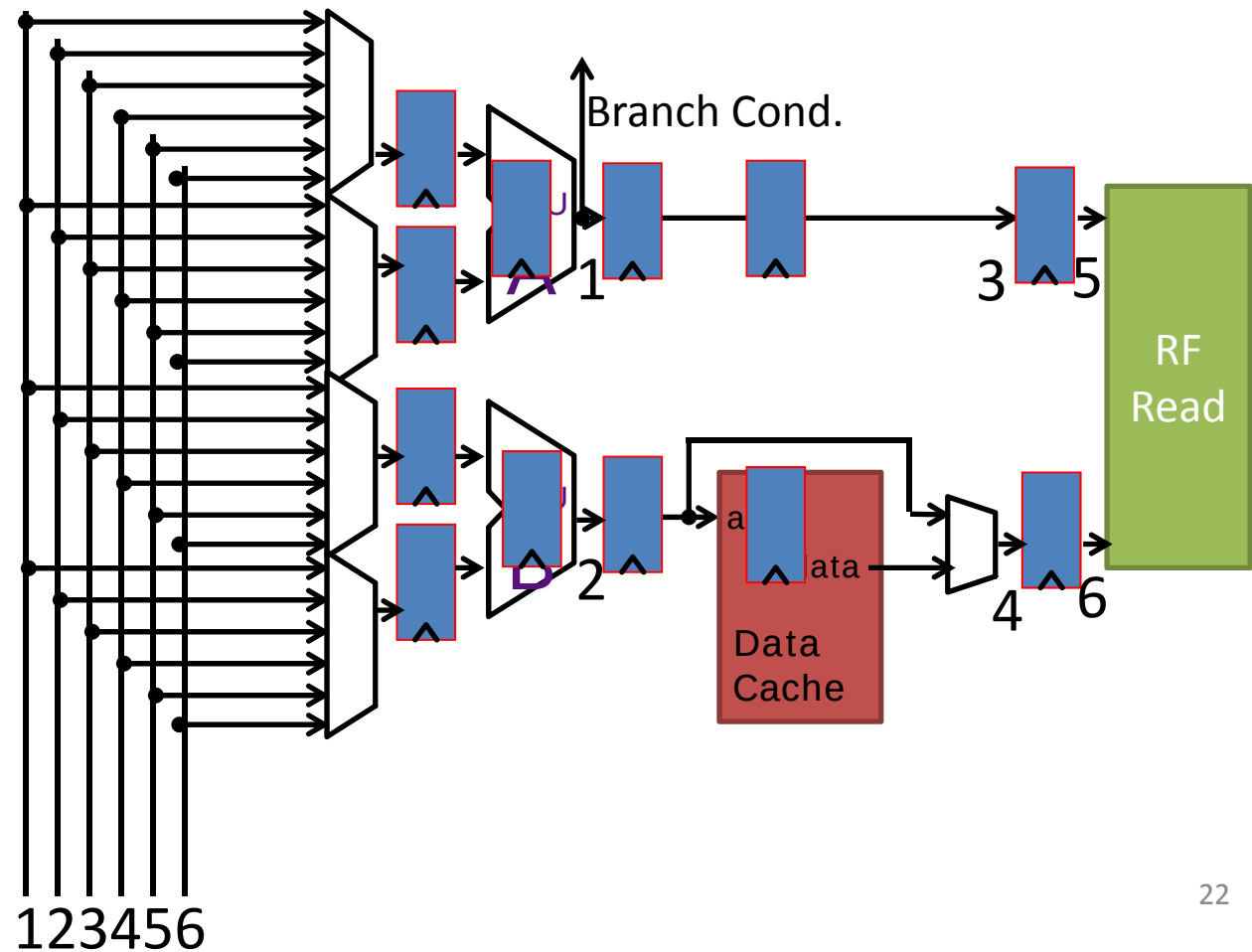
Partitioning/Clustering of Structures

- Faster clock frequency makes global communication challenging
- Faster clock frequency makes centralizes structures harder to meet timing (Ex: Issue Queue, ROB)
- Solution, Partition/Cluster resources and add extra latency to cross between clusters. (Ex 21264)

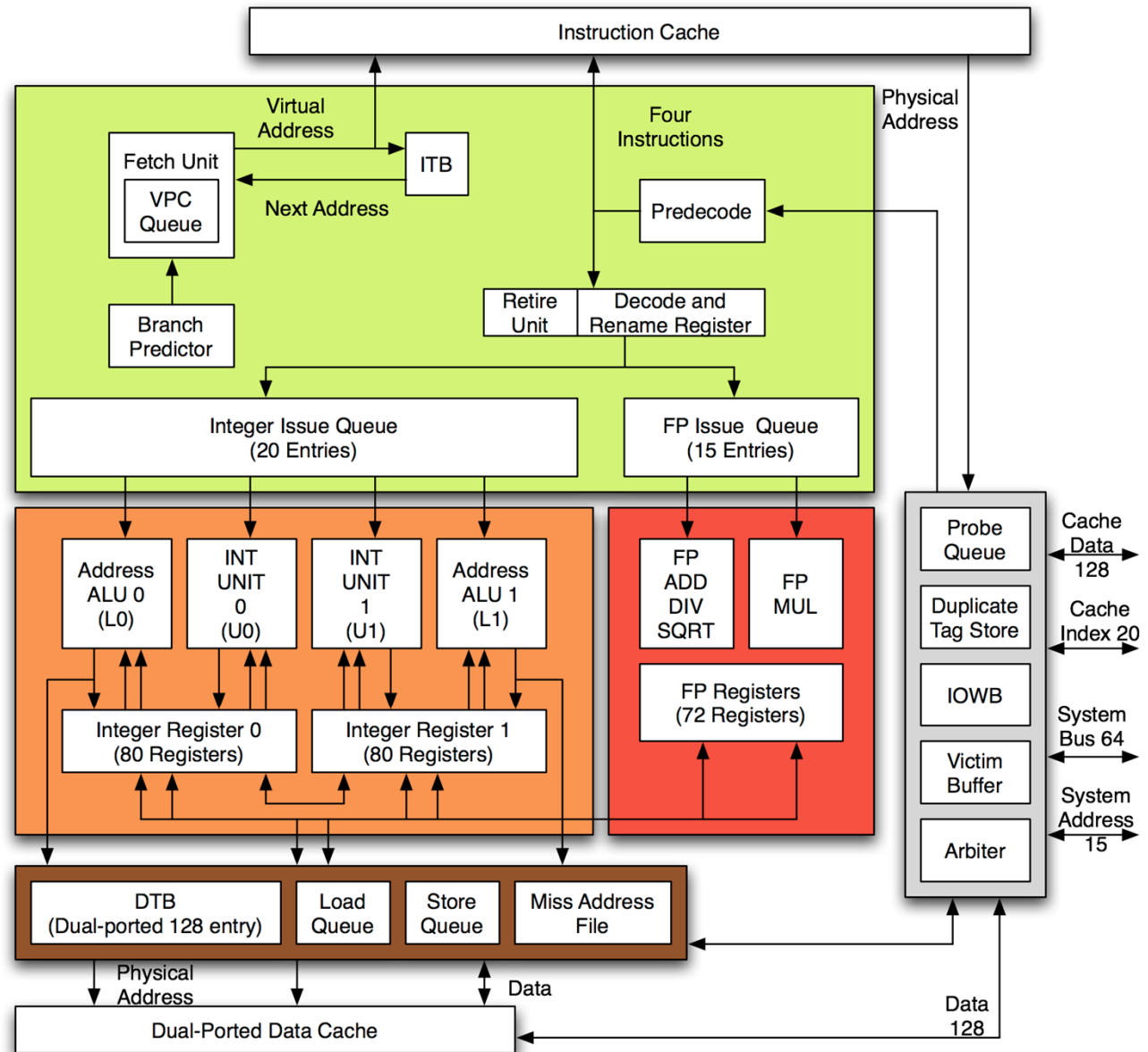
Bypassing in Superscalar Pipelines



Bypassing in Superscalar Pipelines

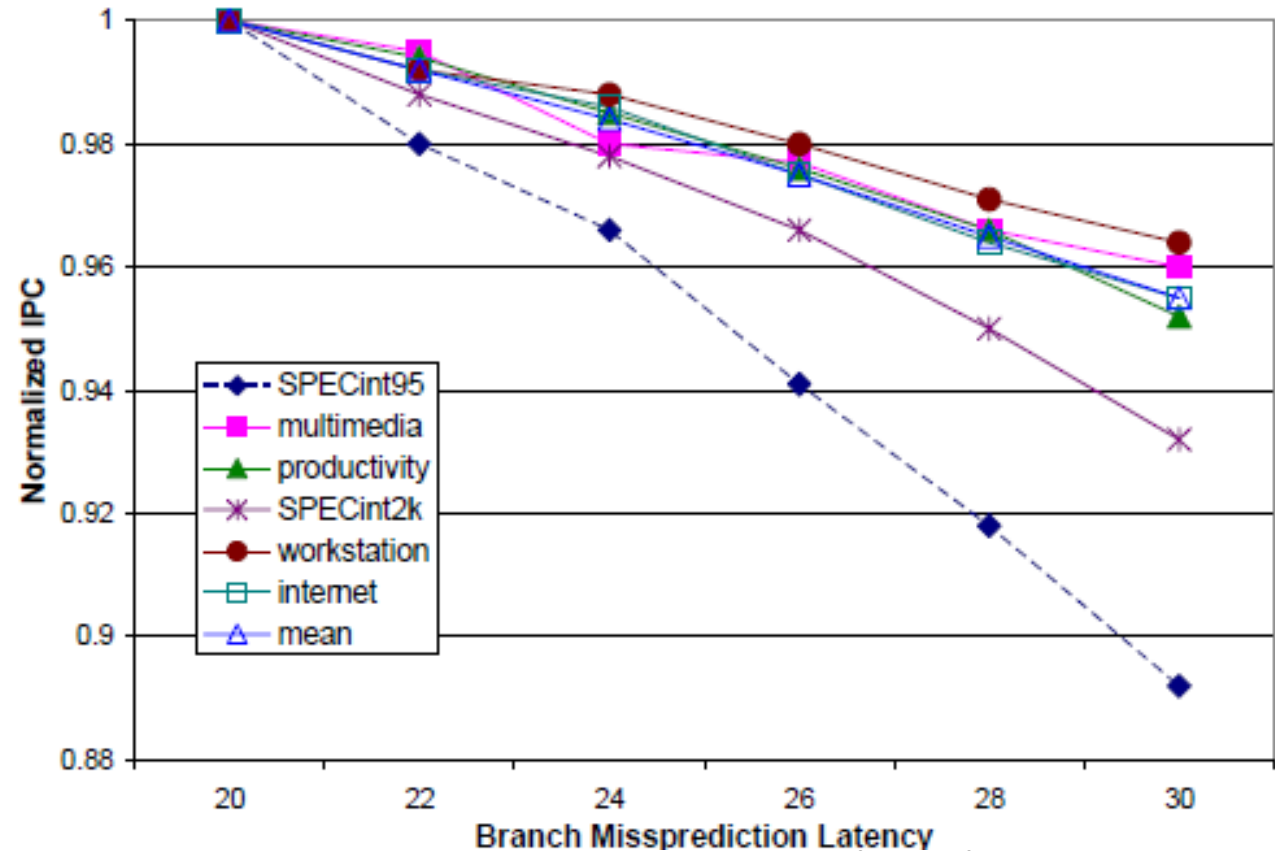


Alpha 21264 Is Clustered



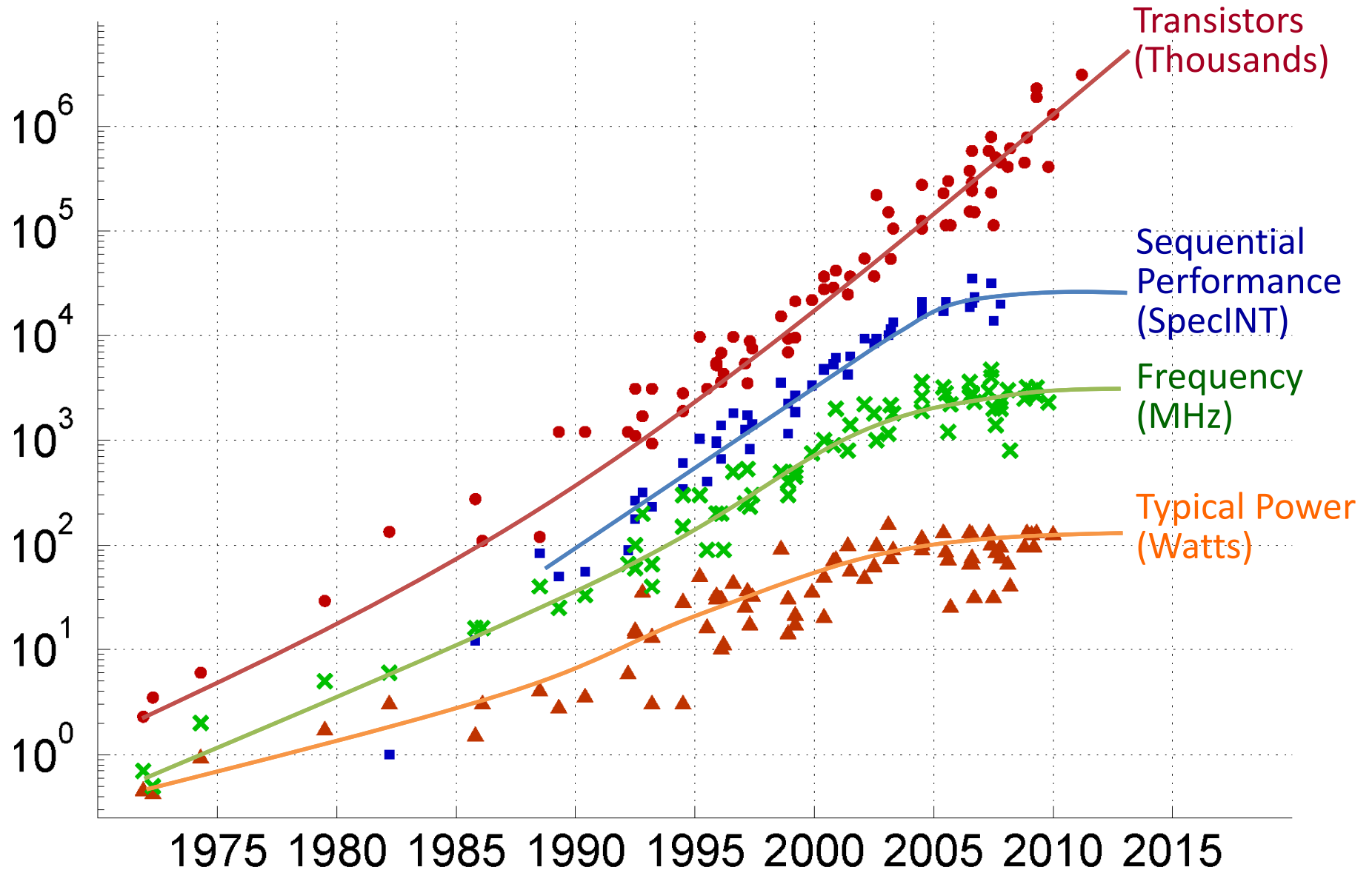
Branch Prediction

- Long Pipes demand spectacular branch predictors



Sprangle and Carmean ISCA 2002

Power Limits Performance



Data collected by M. Horowitz, F. Labonte, O. Shacham, K. Olukotun, L. Hammond, C. Batten, and D. Wentzlaff²⁵

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Acknowledgements

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